

Aperture — Low-Level Design

Module: Administration

Version: v1.0

Module Owner: Platform Operations

Dependencies:

- Authentication Module
 - Users Module
 - Metadata Module
 - Reviews Module
 - Lists Module
 - Discovery Module
 - Streaming Module
 - Notifications Module
 - Analytics Module (future)
-

1. Overview

The Administration module provides operational capabilities for managing and maintaining the Aperture platform.

Unlike user-facing modules, Administration exists to support platform health, content quality, moderation, and operational workflows.

Administrative functionality should remain completely isolated from public APIs wherever possible.

The module is intentionally designed as an operational interface rather than a business domain.

2. Design Goals

The Administration module should:

- Maintain platform integrity
- Enable efficient moderation
- Support operational workflows
- Minimize direct database intervention

- Provide observability into platform health
 - Support future automation
-

3. Module Ownership

Owns

- Administrative actions
 - Moderation workflows
 - Content management
 - User management
 - Operational dashboards
 - Audit records
 - Platform configuration
-

Reads

- Every business module
 - System metrics
 - Logs
 - Analytics
-

Never Modifies Directly

Business rules owned by other modules.

Administration should invoke public service interfaces instead of bypassing them through direct database modifications.

This preserves consistency and auditability.

4. Responsibilities

The Administration module manages:

Users

- View users
- Suspend accounts

- Reinstate accounts
 - Verify accounts (future)
-

Content

- Trigger metadata refresh
 - Hide titles (administrative)
 - Manage featured collections
-

Reviews

- Review reports
 - Remove abusive content
 - Restore moderated content
-

Lists

- Moderate public lists
 - Remove inappropriate collections
-

Platform

- Feature flags
 - Scheduled jobs
 - Cache management
 - System announcements
-

Operations

- Dashboard
 - Health monitoring
 - Job monitoring
 - Deployment status (future)
-

5. High-Level Workflow

Administrative Action



Authentication



Authorization



Validation



Target Service



Audit Log



Response

Every administrative operation should be recorded.

6. Data Flow

Administrator



Admin Portal



Admin API



Authorization



Business Service

↓

Audit Service

↓

Database

↓

Response

Administrative requests should always traverse the same service boundaries as public requests.

7. Administrative Roles

Initial roles:

Administrator

Full platform access.

Moderator

Content moderation only.

Future roles:

- Support
- Content Editor
- Analytics Viewer
- Operations Engineer

Permissions should be role-based rather than endpoint-based.

8. Audit Logging

Every administrative action should generate an audit record.

Examples:

- User suspended
- Review removed
- Metadata refreshed
- Feature flag changed

Audit entries should include:

- actor
- timestamp
- action
- target
- outcome

Audit logs should be immutable.

9. Moderation

Moderation workflows should support:

- Report review
- Report list
- Review queue
- Moderation history

Future additions:

- AI-assisted moderation
- Spam detection
- Reputation scoring

Moderation decisions should remain reviewable.

10. Platform Configuration

Configuration examples:

- Feature flags
- Maintenance mode
- Recommendation weights
- Search tuning
- Notification defaults

Configuration changes should not require code deployments where practical.

11. Background Job Management

Administrators should be able to:

- View scheduled jobs
- Retry failed jobs
- Trigger metadata synchronization
- Trigger search reindexing
- Monitor recommendation generation

Manual intervention should be exceptional rather than routine.

12. Operational Dashboards

Future dashboards include:

- User growth
- Review activity
- Search volume
- Recommendation performance
- Background job status
- Cache health
- API latency

Dashboards should surface trends rather than raw logs.

13. Security Considerations

Administrative functionality requires elevated privileges.

Security measures include:

- Role-based access control
- Audit logging
- Session expiration
- Re-authentication for sensitive actions
- Least privilege principle

Administrative endpoints should never be exposed through public routing.

14. Performance Considerations

Administrative traffic is expected to be low.

Optimization priorities:

- Reliability
- Auditability
- Correctness

over raw throughput.

Long-running administrative tasks should execute asynchronously where appropriate.

15. Error Handling

Representative failures:

- Unauthorized access
- Insufficient permissions
- Invalid administrative action
- Target resource unavailable
- Background job failure

Errors should be logged with sufficient operational detail while exposing only safe information to administrators.

16. Testing Strategy

Unit Tests

- Permission evaluation
 - Role validation
 - Audit creation
-

Integration Tests

- Administrative workflows
- Moderation actions
- Job management

API Tests

- Role enforcement
- Administrative endpoints
- Feature configuration

Security Tests

- Privilege escalation
- Unauthorized access
- Audit completeness

17. Operational Considerations

Key operational metrics:

- Moderation queue length
- Failed background jobs
- Administrative login attempts
- Audit log growth
- Feature flag usage

Failures in the Administration module should not impact public user functionality.

18. Future Service Extraction

The Administration module is **not** expected to become an independent microservice.

Reasons:

- Low traffic
- Tight integration with operational workflows
- Cross-domain visibility requirements

However, supporting systems such as analytics, moderation, or audit logging may eventually evolve into independent services if operational complexity increases.

19. Future Evolution

Potential enhancements:

- AI-assisted moderation
- Automated abuse detection
- Content quality scoring
- Bulk operations
- Multi-tenant administration
- Operational runbooks
- Incident management integration
- Live system health dashboards

The Administration module should remain focused on operational tooling rather than business logic.

20. Interview Discussion Points

Be prepared to explain:

- Why isolate administrative functionality from business modules?
 - Why should administrative actions flow through service interfaces instead of directly modifying the database?
 - Why are audit logs immutable?
 - How would you implement fine-grained role-based access control?
 - Why is Administration a poor candidate for service extraction?
 - How would you safely introduce feature flags into the platform?
-

21. Summary

The Administration module provides the operational capabilities required to run Aperture as a production platform.

By centralizing moderation, configuration, auditing, and operational workflows while respecting the boundaries of other business modules, the module enables maintainable platform operations without compromising the architecture or bypassing established domain rules.